INCH-POUND
MIL-G-63343 (AR)
AMENDMENT 4
10 July 1998
SUPERSEDING
AMENDMENT 3
4 January 1994

MILITARY SPECIFICATION

GRENADE, M74, HEMISPHERES FOR

MIL-G-63343 was inactivated after 13 February 1996 for new design.

This amendment forms a part of Military Specification MIL-G-63343(AR), dated 11 July 1980, and is approved for use by the U.S. Army Armament Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

Change item designation from "XM74" to "M74" in title and wherever else it appears in this specification.

PAGE 1

2.1 Delete "MIL-I-6866" and substitute "MIL-STD-6866".

Delete "FSC 1375" and substitute "FSC 1330".

PAGE 3

3.2.1.1 Microstructure. Delete paragraph in its entirety and substitute the following:

"3.2.1.1 Microstructure. The microstructure shall exhibit a uniform distribution of tungsten and binder material, and matrix depletion (porosity) shall not exceed 0.15% (volume) when viewed at 200 magnifications and tested as specified in 4.5.3."

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4.4.2.1 Delete Minors 201 and 202 in their entirety.

Minor 209: Delete in its entirety.

Minor 211, Add the following: "Straighness incorrect (2 places 0.65%, 3.2, Gage".

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4.5.3 Microstructure. Delete paragraph in its entirety and substitute the following:

"4.5.3 Microstructure. The test shall be performed on a test blank from a disc using any method satisfactory to the procuring agency. (The measurement techniques outlined in Metals Handbook, 9th Edition, Vol. 9, A.S.M., are approved and recommended.) Only the following will be counted for the purpose of porosity value measurement:

a. all areas of binder depletion.
b. sub-surface blisters.
c. porosity caused by inclusions with a dimension greater than 10 microns.
d. shrinkage breaks in the binder or gamma phase having a length greater than 10 microns.

If a number of failures exceed the requirments of 4.4.2, the lot shall be rejected."

* 4.5.4 Mechanical properties. Delete paragraph in its entirety and substitute:

"4.5.4 Mechanical properties. Tensile test blanks conforming to ASTM E8, Figure 6, shall be prepared from completed discs except that dimension C may be .630", length to be outside diameter of sintered disc and a hole .250" + .005 diameter may be drilled in each grip section to aid in test.

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These samples shall be free of scoring when subjected to testing at a strain rate no greater than .10 inch per inch per minute as specified in ASTM E8. If any test specimen fails to meet ultimate tensile strength and elongation requirements, it shall be classified defective. If the number of failures exceeds the requirements of 4.4.2, the lot shall be rejected."

4.5.5 Penetrant inspection. First sentence, Delete "MIL-I-6866, Table I" and substitute "MIL-STD-6866".

6.3 Submission of inspection equipment for design approval. Delete last sentence in its entirety and substitute: "Submit designs as required to "Commander, US Army Armament, Research, Development and Engineering Center, ATTN: AMSTA-AR-QAA-B, Picatinny Arsenal, NJ 07806-5000."

6.4 Informational tests. Delete paragraph in its entirety.

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in this notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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